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Semester: III	**Year : AY 2026-27**
Subject: Programming with Java	
ASSIGNMENT No :	
TITLE : JDBC ResultSet	

Problem Statement

Develop a Java application using JDBC to navigate and retrieve records from a ResultSet.

Learning Objectives

To navigate and retrieve records using ResultSet methods.

Exercise

1. **Create a program to display database records one by one using ResultSet.**

Code:

```
import java.sql.*;

public class StudentRecords {

   public static void main(String[] args) {

      String url = "jdbc:mysql://localhost:3306/college";
      String user = "root";
      String password = "mini@12";

      try {
         Class.forName("com.mysql.cj.jdbc.Driver");

         Connection con = DriverManager.getConnection(url, user, password);

         String query = "SELECT * FROM student";

         Statement stmt = con.createStatement();
```

```
            ResultSet rs = stmt.executeQuery(query);

            System.out.println("Student Records:");

            while (rs.next()) {
                System.out.println(
                    rs.getInt("roll_no") + " | " +
                    rs.getString("name") + " | " +
                    rs.getString("course") + " | " +
                    rs.getDouble("marks")
                );
            }

            con.close();

        } catch (Exception e) {
            System.out.println(e);
        }
    }
}
```

Output:

```
rogram Files\Java\jdk-27\bin\java.exe' '-agentlib:
jdwp=transport=dt_socket,server=n,suspend=y,addres
s=localhost:59831' '@C:\Users\minis\AppData\Local\
Temp\cp_7p6f0nbkg00gfn7benzo3fpbn.argfile' 'Studen
tRecords'
Student Records:
101 | Rahul | CSE | 85.0
102 | Priya | CSE | 90.0
PS C:\Users\minis\Downloads\javaprograms>
```

2. **Create a Java program that retrieves employee records from a database and displays them one by one using ResultSet. Show employee ID, name, department, and salary sequentially**

Code:

```
import java.sql.*;

public class EmployeeRecords {

    public static void main(String[] args) {
```

```
        String url = "jdbc:mysql://localhost:3306/college";
        String user = "root";
        String password = "mini@12";

        try {
            Class.forName("com.mysql.cj.jdbc.Driver");

            Connection con = DriverManager.getConnection(url, user, password);

            String query = "SELECT * FROM employee";

            Statement stmt = con.createStatement();

            ResultSet rs = stmt.executeQuery(query);

            System.out.println("Employee Records:");

            while (rs.next()) {
                System.out.println(
                    rs.getInt("employee_id") + " | " +
                    rs.getString("name") + " | " +
                    rs.getString("department") + " | " +
                    rs.getDouble("salary")
                );
            }

            con.close();

        } catch (Exception e) {
            System.out.println(e);
        }
    }
}
```

Output:

```
jdwp=transport=dt_socket,server=n,suspend=y,addres
s=localhost:55907' '@C:\Users\minis\AppData\Local\
Temp\cp_7p6f0nbkg00gfn7benzo3fpbn.argfile' 'Employ
eeRecords'
Employee Records:
201 | Amit | HR | 45000.0
202 | Sneha | IT | 55000.0
203 | Rohan | Finance | 50000.0
PS C:\Users\minis\Downloads\javaprograms>
```

Conclusion

In this practical, we learned how to use the ResultSet interface in JDBC to retrieve and display records from a database. We successfully displayed student and employee records one by one using the rs.next() method. We also used ResultSet methods to retrieve different types of data such as integers, strings, and double values. This helped us understand how Java navigates through database records and displays the required information seque